Joachim Bona-Pellissier

Curriculum Vitae

MaLGa Center
Università di Genova
Via Dodecaneso 35
16146 Genova, Italy
joachim.bona@edu.unige.it
joachimbp.github.io

Current position

2023 – present **Postdoctoral researcher**, *Machine Learning Genoa Center (MaLGa), Università degli Studi di Genova*, Italy.
Under the supervision of Prof. Lorenzo Rosasco.
Topics of interest: Physics-informed machine learning, Kernel methods, Deep learning theory, Geometry of ReLU networks.

Education

2019 – 2023 **Ph.D. in Applied Mathematics**, *Institut de Mathématiques de Toulouse, ANITI (Artificial and Natural Intelligence Toulouse Institute), Toulouse 1 Capitole University*, Toulouse, France.
Under supervision of Prof. François Malgouyres and Prof. François Bachoc.
Subject: On the Identifiability of Deep ReLU Neural Networks.
Commitee: Francis Bach, Gérard Biau, Rémi Gribonval (reviewer), Elisenda Grigsby (reviewer), Edouard Pauwels (president).

2018 – 2019 **M.Sc. in Mathematics, Vision and Learning (MVA) (with high honors)**, *ENS Paris-Saclay.*

2017 – 2018 **Agrégation in Mathematics (national rank: 47)**, *ENS Paris-Saclay*, Probability and Statistics minor.

2016 – 2017 **Master's degree in Fundamental Mathematics (with high honors)**, *Grenoble Alpes University.*
Research work on the Berkovich disk (p-adic analysis), supervised by Andrea Pulita.

2015 – 2016 **B.Sc. in Fundamental Mathematics**, *ENS de Lyon.*

2012 – 2015 **Math & Physics undergrad studies (prépa MPSI - MP*)**, *Lycée Champollion*, Grenoble.

Preprints

2026 **PIKS: Universal Physics-Informed Kernel Methods**
Joachim Bona-Pellissier, Giacomo Meanti, Matteo Santacesaria, Lorenzo Rosasco.

2024 **Geometry-induced Regularization in Deep ReLU Neural Networks**
Joachim Bona-Pellissier, Francois Malgouyres, Francois Bachoc.

Publications

2026 **Fast Learning Rates for Physics-Informed Kernel Methods**
Luc Brogat-Motte, Joachim Bona-Pellissier, Giacomo Meanti, Lorenzo Rosasco.
Accepted at NeurIPS 40.

2023 **Parameter identifiability of a deep feedforward ReLU neural network**
Joachim Bona-Pellissier, François Bachoc, and François Malgouyres.
In: Machine Learning, 112(11), 4431-4493.

2022 **Local Identifiability of Deep ReLU Neural Networks: the Theory**
Joachim Bona-Pellissier, François Malgouyres and François Bachoc.
In: NeurIPS, Thirty-Sixth Conference on Neural Information Processing Systems.

Talks

2026 **Minisymposium "Theory-guided Design of Deep Learning-based Surrogates"**, *WCCM ECCOMAS 2026*, Munich, Germany.

2026 **InterCoML Seminar**, *COST Action InterCoML* , Online.

2026 **MATH4AIML - UMI Workshop**, *Sapienza Università di Roma* , Italy.

2025 **Représentations Neuronales Implicites : des NeRF aux PINN**, *Institut Henri Poincaré*, Paris, France.
"Universal physics-informed machine learning with kernels."

2025 **NPTwins**, *Università di Messina*, Italy.
"Universal physics-informed machine learning with kernels."

2024 **MML seminar**, *Max Planck Institute MIS + UCLA*, Online.
"Geometry-induced Implicit Regularization in Deep ReLU Neural Networks."

2022 **3IA Doctoral Workshop**, *MIAI institute, Université Grenoble Alpes*, Grenoble, France.

2022 **Research school "Statistical and geometric discrepancies for machine learning"**, *Centre Henri Lebesgue*, Rennes, France.

Mentoring

2026 **Florentin Acker**, *research intern at MaLGa*, master level (M1).

2026 **Hubert Leroux**, *research intern at MaLGa*, master level (M1).

2025 **Mathis Guichet**, *research intern at MaLGa*, master level (M2).

2024 **Rayan Autones**, *research intern at MaLGa*, master level (M1)

Internships

2019 **Research intern at IRT Saint-Exupéry**, Toulouse, supervised by François Malgouyres and Grégory Flandin.
I worked with the DEEL team on the question of identifiability and interpretability of neural networks.

2016 **Research intern at Aix-Marseille Université**, supervised by Guillemette Chapuisat.
I worked on the modeling of tumor heterogeneity for cancer treatments.

Teaching

2022 – 2023 **Deep Learning practical**, *Graduate (mathematics)*, Paul Sabatier University, Toulouse.
Introduction to MLPs, CNNs, Autoencoders, U-nets... Introduction to Keras, Tensorflow, Scikit-learn.

2022 – 2023 **Optimization practical**, *Graduate (mathematics)*, Paul Sabatier University, Toulouse.
First order methods, Newton/quasi-Newton methods. Quadratic problems, SQP method.

2022 – 2023 **Machine Learning tutorial & practical**, *Undergraduate (mathematics)*, Paul Sabatier University, Toulouse. Linear regression, ridge and LASSO penalization, KNN, logistic regression, SVM, K-means.

2022 – 2023 **Linear Algebra tutorial**, *Undergraduate (mathematics)*, Paul Sabatier University, Toulouse.
Euclidian and hermitian spaces, isometries, spectral theorem, quadratic forms, and more.

2021 – 2022 **Math oral examinations**, *Undergraduate (mathematics)*, Paul Sabatier University, Toulouse.
Topology, Algebra, Differential calculus.

2020 – 2022 **Convex Optimization tutorial (in English)**, *Graduate (finance)*, Toulouse School of Management.
Optimization without or with constraints, Lagrange Theorem, KKT Conditions.

2019 – 2022 **Methods of math. analysis tutorial**, *Undergraduate (management and business administration)*, Toulouse 1 Capitole University. Introduction to Probabilities, Statistics and Financial Mathematics.

2019 – 2020 **Math oral examinations**, *Undergraduate (mathematics and social sciences)*, Toulouse School of Economics. Introduction to Series, Probability, Integral Calculus.

2018 – 2019 **Math oral examinations**, *Undergraduate (Classes prépa MP)*, Lycée Fénelon Sainte-Marie, Paris.
Algebra, Topology, Integral Calculus, Differential Calculus, Functional analysis, ODEs, Probability.

2016 – 2017 **Student mentoring**, *3rd year bachelor (L3) in mathematics*, Grenoble Alpes University.

Languages

French	Native language
Spanish	Fluent
English	Fluent
Italian	Proficient